```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report


# 1. LOAD DATA

df = pd.read_csv("combined_data.csv")

# ==============================
# 2. CLEAN COLUMN NAMES
# ==============================
df.columns = df.columns.str.strip()

# ==============================
# 3. HANDLE MISSING VALUES
# ==============================
df["Returns"] = df["Returns"].fillna(0)

# ==============================
# 4. CONVERT DATE
# ==============================
df["Purchase Date"] = pd.to_datetime(df["Purchase Date"])

# Create new useful features from date
df["Year"] = df["Purchase Date"].dt.year
df["Month"] = df["Purchase Date"].dt.month
df["Day"] = df["Purchase Date"].dt.day

# ==============================
# 5. DROP UNNECESSARY COLUMNS
# ==============================
df = df.drop(columns=["Customer Name"])  # name ML me useful nahi hota

# Agar Customer Age aur Age dono hain to ek remove karo
if "Customer Age" in df.columns:
    df = df.drop(columns=["Customer Age"])

# ==============================
# 6. ENCODE CATEGORICAL DATA
# ==============================
le = LabelEncoder()

df["Product Category"] = le.fit_transform(df["Product Category"])
df["Payment Method"] = le.fit_transform(df["Payment Method"])
df["Gender"] = le.fit_transform(df["Gender"])

# ==============================
# 7. DEFINE FEATURES & TARGET
# ==============================
```

```
X = df[[
    "Product Category",
    "Product Price",
    "Quantity",
    "Total Purchase Amount",
    "Payment Method",
    "Age",
    "Gender",
    "Year",
    "Month"
]]

y = df["Churn"]

# ================================
# 8. TRAIN TEST SPLIT
# ================================
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ================================
# 9. TRAIN MODEL
# ================================
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# ================================
# 10. PREDICTION
# ================================
y_pred = model.predict(X_test)

# ================================
# 11. RESULTS
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
```